MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH & STUDIES

SCHOOL OF COMPUTER APPLICATIONS

PRACTICAL FILE

Data Structure and Algorithm

Academic Session: 2026–2027

SUBMITTED TO

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MRIIRS

SUBMITTED BY

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Course: MCA

Semester: I

```python
arr = [10, 20, 30, 40, 50]

n = 5

print("Original array:", arr)

element = int(input("Enter element to delete: "))

pos = -1
i = 0

while i < n:
    if arr[i] == element:
        pos = i
        break
    i = i + 1

if pos == -1:
    print("Element not found")

else:
    i = pos

    while i < n - 1:
        arr[i] = arr[i + 1]
        i = i + 1

    n = n - 1

    print("Array after deletion:")

    i = 0
    while i < n:
        print(arr[i], end=" ")
        i = i + 1
```

```
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code> & c:\py
Original array: [10, 20, 30, 40, 50]
Enter element to delete: 30
Array after deletion:
10 20 40 50
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

```python
arr = [10, 20, 30, 40, 50, 0]

n = 5

print("Original array:", arr[:n])

element = int(input("Enter element to insert: "))
pos = int(input("Enter position (1 to 6): "))

if pos < 1 or pos > n + 1:
    print("Invalid position")

else:
    i = n

    while i >= pos:
        arr[i] = arr[i - 1]
        i = i - 1

    arr[pos - 1] = element
    n = n + 1

    print("Array after insertion:")
    i = 0

    while i < n:
        print(arr[i], end=" ")
        i = i + 1
```

```
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code> & c:\pyt
Original array: [10, 20, 30, 40, 50]
Enter element to insert: 25
Enter position (1 to 6): 2
Array after insertion:
10 25 20 30 40 50
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

```python

students = []
n = 5
subjects = 3

i = 0

while i < n:
    print("Enter marks for Student", i + 1)

    marks = []
    j = 0
    total = 0

    while j < subjects:
        mark = float(input("Enter subject marks: "))

        marks = marks + [mark]
        total = total + mark

        j = j + 1

    percentage = (total / 300) * 100

    students = students + [[total, percentage]]

    i = i + 1

print("\nStudent Results")

i = 0
while i < n:
    print("Student", i + 1)
    print("Total Marks:", students[i][0])
    print("Percentage:", students[i][1], "%")

    i = i + 1

```

```
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code> & c:\python314\
Enter subject marks: 89
Enter subject marks: 34
Enter subject marks: 23
Enter marks for Student 4
Enter subject marks: 89
Enter subject marks: 98
Enter subject marks: 78
Enter marks for Student 5
Enter subject marks: 89
Enter subject marks: 67
Enter subject marks: 56

Student Results
Student 1
Total Marks: 212.0
Percentage: 70.66666666666667 %
Student 2
Total Marks: 202.0
Percentage: 67.33333333333333 %
Student 3
Total Marks: 146.0
Percentage: 48.66666666666667 %
Student 4
Total Marks: 265.0
Percentage: 88.33333333333333 %
Student 5
Total Marks: 212.0
Percentage: 70.66666666666667 %
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

```
r1 = int(input("Enter rows of first matrix: "))
c1 = int(input("Enter columns of first matrix: "))

r2 = int(input("Enter rows of second matrix: "))
c2 = int(input("Enter columns of second matrix: "))

if c1 != r2:
    print("Matrix multiplication is not possible")

else:
    A = []
    B = []
    C = []

    print("Enter first matrix elements:")

    i = 0
    while i < r1:
        row = []
        j = 0

        while j < c1:
            value = int(input())
            row = row + [value]
            j = j + 1

        A = A + [row]
        i = i + 1

    print("Enter second matrix elements:")

    i = 0
    while i < r2:
        row = []
        j = 0

        while j < c2:
            value = int(input())
            row = row + [value]
            j = j + 1

        B = B + [row]
        i = i + 1

    i = 0
    while i < r1:
        row = []
        j = 0
```

```
        j = 0

        while j < c2:
            total = 0
            k = 0

            while k < c1:
                total = total + A[i][k] * B[k][j]
                k = k + 1

            row = row + [total]
            j = j + 1

        C = C + [row]
        i = i + 1

    print("Product of matrices:")

    i = 0
    while i < r1:
        j = 0

        while j < c2:
            print(C[i][j], end=" ")
            j = j + 1

        print()
        i = i + 1
```

```
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code> & c:\python
Enter rows of first matrix: 2
Enter columns of first matrix: 2
Enter rows of second matrix: 2
Enter columns of second matrix: 2
Enter first matrix elements:
2
3
5
6
Enter second matrix elements:
8
9
5
6
Product of matrices:
31 36
70 81
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

```python
n = int(input("Enter size of square matrix: "))

matrix = []

print("Enter matrix elements:")

for i in range(n):
    row = []
    for j in range(n):
        element = int(input())
        row.append(element)
    matrix.append(row)

print("Matrix:")
for i in range(n):
    for j in range(n):
        print(matrix[i][j], end=" ")
    print()

diagonal_sum = 0
non_diagonal_sum = 0

for i in range(n):
    for j in range(n):
        if i == j:
            diagonal_sum = diagonal_sum + matrix[i][j]
        else:
            non_diagonal_sum = non_diagonal_sum + matrix[i][j]

print("Sum of diagonal elements:", diagonal_sum)
print("Sum of non-diagonal elements:", non_diagonal_sum)
```

```
Enter size of square matrix: 2
Enter matrix elements:
5
6
3
2
Matrix:
5 6
3 2
Sum of diagonal elements: 7
Sum of non-diagonal elements: 9
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

Users > asus > Desktop > DSA_python Shens > Queue > general.py > ...

```python
size = int(input("Enter queue size: "))

queue = []
front = -1
rear = -1

while True:
    print("\n--- QUEUE OPERATIONS ---")
    print("1. Enqueue (Insert)")
    print("2. Dequeue (Delete)")
    print("3. Display")
    print("4. Peek")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if rear == size - 1:
            print("Queue Overflow")

        else:
            element = int(input("Enter element: "))

            if front == -1:
                front = 0

            queue = queue + [element]
            rear = rear + 1

            print("Element inserted successfully")

    elif choice == 2:
        if front == -1 or front > rear:
            print("Queue Underflow")

        else:
            print("Deleted element:", queue[front])
            front = front + 1

            if front > rear:
                front = -1
                rear = -1
                queue = []

    elif choice == 3:
        if front == -1:
            print("Queue is empty")

        else:
            print("Queue elements:")

            i = front
            while i <= rear:
                print(queue[i], end=" ")
                i = i + 1

            print()

    elif choice == 4:
        if front == -1:
            print("Queue is empty")
```

```
58
59      elif choice == 4:
60          if front == -1:
61              print("Queue is empty")
62
63          else:
64              print("Front element:", queue[front])
65
66      elif choice == 5:
67          print("Program terminated")
68          break
69
70      else:
71          print("Invalid choice")
72
```

```
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe "c:/Users/asus/Desktop/DSA_pyht
Enter queue size: 3

--- QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element: 2
Element inserted successfully

--- QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element: 5
Element inserted successfully

--- QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 3
Queue elements:
2 5

--- QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 2
Deleted element: 2

--- QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 5
Program terminated
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

```python

size = int(input("Enter circular queue size: "))

queue = [0] * size
front = -1
rear = -1

while True:
    print("\n--- CIRCULAR QUEUE OPERATIONS ---")
    print("1. Enqueue (Insert)")
    print("2. Dequeue (Delete)")
    print("3. Display")
    print("4. Peek")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if (rear + 1) % size == front:
            print("Circular Queue Overflow")

        else:
            element = int(input("Enter element: "))

            if front == -1:
                front = 0
                rear = 0
            else:
                rear = (rear + 1) % size

            queue[rear] = element
            print("Element inserted successfully")

    elif choice == 2:
        if front == -1:
            print("Circular Queue Underflow")

        else:
            print("Deleted element:", queue[front])

            if front == rear:
                front = -1
                rear = -1
            else:
                front = (front + 1) % size

    elif choice == 3:
        if front == -1:
            print("Circular Queue is empty")

        else:
            print("Circular Queue elements:")

            i = front

            while True:
                print(queue[i], end=" ")
```

```
elif choice == 3:
    if front == -1:
        print("Circular Queue is empty")

    else:
        print("Circular Queue elements:")

        i = front

        while True:
            print(queue[i], end=" ")

            if i == rear:
                break

            i = (i + 1) % size

        print()

elif choice == 4:
    if front == -1:
        print("Circular Queue is empty")

    else:
        print("Front element:", queue[front])

elif choice == 5:
    print("Program terminated")
    break

else:
    print("Invalid choice")
```

```
Enter circular queue size: 3

--- CIRCULAR QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element: 2
Element inserted successfully

--- CIRCULAR QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element: 5
Element inserted successfully

--- CIRCULAR QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 2
Deleted element: 2

--- CIRCULAR QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 3
Circular Queue elements:
5

--- CIRCULAR QUEUE OPERATIONS ---
1. Enqueue (Insert)
2. Dequeue (Delete)
3. Display
4. Peek
5. Exit
Enter your choice: 5
Program terminated
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

```python
size = int(input("Enter stack size: "))
stack = []
top = -1

while True:
    print("\n--- STACK OPERATIONS ---")
    print("1. Push")
    print("2. Pop")
    print("3. Display")
    print("4. Peek")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if top == size - 1:
            print("Stack Overflow")

        else:
            element = int(input("Enter element to push: "))
            stack = stack + [element]
            top = top + 1
            print("Element pushed successfully")

    elif choice == 2:
        if top == -1:
            print("Stack Underflow")

        else:
            print("Popped element:", stack[top])
            stack = stack[:top]
            top = top - 1

    elif choice == 3:
        if top == -1:
            print("Stack is empty")

        else:
            print("Stack elements (Top to Bottom):")
            i = top

            while i >= 0:
                print(stack[i])
                i = i - 1

    elif choice == 4:
        if top == -1:
            print("Stack is empty")

        else:
            print("Top element:", stack[top])

    elif choice == 5:
        print("Program terminated")
        break

    else:
        print("Invalid choice")
```

```
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe "c:/Users/asus/De
Enter stack size: 5

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element to push: 2
Element pushed successfully

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element to push: 3
Element pushed successfully

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 2
Popped element: 3

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 3
Stack elements (Top to Bottom):
2

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 5
Program terminated
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

```python
string = input("Enter a string: ")

stack = []
top = -1

for ch in string:
    stack = stack + [ch]
    top = top + 1


reverse = ""

while top >= 0:
    reverse = reverse + stack[top]
    top = top - 1

print("Original string:", string)
print("Reversed string:", reverse)
```

```
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code> & c:\p
Enter a string: sneha kumari
Original string: sneha kumari
Reversed string: iramuk ahens
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```

```python
expression = input("Enter an expression: ")

stack = []
top = -1
balanced = True

for ch in expression:

    if ch == '(':
        stack = stack + [ch]
        top = top + 1

    elif ch == ')':
        if top == -1:
            balanced = False
            break

        else:
            top = top - 1
            stack = stack[:top + 1]

if top != -1:
    balanced = False

if balanced:
    print("Parentheses are balanced")
else:
    print("Parentheses are not balanced")
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code> & c:
Enter an expression: {[(a+b)*(c-d)] + ((x^2+y^2)/(m-n))}
Parentheses are balanced
PS C:\Users\asus\AppData\Local\Programs\Microsoft VS Code>
```